Muhd Lai Razi Muhd Silmi

+673 815 3331 / +60 10 376 2645 · laihere281@gmail.com
NO25, LOT6433, SPG286-30-35-9, RPN LUMUT, MUKIM LIANG

CAREER OBJECTIVE

A Bachelor of Computer Science graduate from Monash University Malaysia, keen in data analytics and data science role. Equipped with data exploration and study, data processing, analytical skills in visualization and descriptive manner followed by data automation modeling.

PROFESSIONAL EXPERIENCE

Al-Kindi Coding Centre **April 2026 - Current**
Entry AI/ML Engineer

- Tested and compared document query chat with available **AI models (e.g. gpt, deepseek)**:
  - Input and output token costs
  - Token usage and cost for input and output
  - Model's 'memory' (full context window) on current document conversation
  - Model's response time, accuracy and quality (qualitative/subjective)
- **Structurized** and **normalized** document information extraction pipeline on document's in-depth categories and metadata for insights and visuals (e.g. Invoice, Purchase Orders).
- Generated **insights** and **visualization** (bar chart, pie chart, line-chart) based on document's category and metadata.

Al-Kindi Coding Centre **January 2026 - March 2026**
AI/ML Intern

- Implemented and tested open-sourced **OCR text extraction** with improved text output readability, run-time, resource-usage and content structure.
- Performed **document classification** and **generative tags** with open-sourced **zero-shot classification** model and **tag generator** model from **HuggingFace.**
- Extracted document's metadata information (e.g. author, date) with **NER** (Named Entity Recognition).
- Documents and datasets sample collection from Kaggle for **limit testing performance** on obtained models (accuracy, run-time and resource-usage).

Mossery Sdn Bhd **July 2024 - October 2024**
Supply Chain and Inventory Intern

- Forecast, project, plan and execute on products to be produced and sold based on past years trend and season.
- Implemented **semi-automation dashboard** with the use of **Google Sheets** and **Power BI** to integrate with other departments (logistics, production and fulfillment team) to ensure cohesive work flow and reduce repetitive manual work.
- Performed **further manual analysis** with **R** for further findings regarding raw material purchases, shipping costs followed by products supply and demand in off-season and peak-season.
- Coordinated with other departments (logistics, production and fulfilment team) regarding purchases of raw materials and out-source services to production and fulfilment timeline.
- Prepared and presented the reports and updates from supply chain and inventory department for monthly town hall meetings.

PROJECT

Final Year Project
Generative AI for Humanity

- Researched and implemented two open source **unconditional generative AI** for image generation with its original image dataset.
- Built an open source **face recognition model** for face classification of generated image by the two model into a classified variables dataset.
- Performed data analysis with bias metrics (**Cramer's V** and **Ducher's Z)** from the produced face classification dataset to compare the similarity between original image dataset and generated image dataset by the two implemented generative image model.
- Executed software testing and quality assurance to ensure the project requirements are met and worked as intended.
- Worked on basic application and command prompt to run the models with provided instructions step by step (user guide).
- Contributed and produced a project documentation that includes the research background of given scope (bias or not bias), generative AI models obtained with its dataset, use case, limitation and future improvements.

Data Analytics
1. Covid-19 Pro Social Behavior Questionnaires Study

- Explored the given data "CoronaBaseline" that includes dimension, class, range, mean and null values.
- Performed **data pre-processing** that includes removing unwanted variables (high null values) and removing unwanted entries.
- Implemented **linear model** to obtain top predictor for each "covid pro social behavior" variable value and observed the difference and similarity between given focus country and other countries as a group.
- Researched and gathered external dataset that may correlate with the given dataset "CoronaBaseline" such as vaccination rate, GDP and development index followed by executing **data tidying** and **wrangling**.
- Extracted similar countries with given focus country based on **clustering values** through clean external dataset. Said similar countries are used to observe if same top predictor for each "covid pro social behavior" are similar with or distinct from focus country.

2. More Humid Tomorrow Prediction

- Studied the given data "HumidPredict" that includes dimension, class, range, mean and null values.
- Performed **data pre-processing** that includes removing unwanted variables (non-categorical, non-numeric or mostly null), removing unwanted entries, convert categorical string variables value into simpler terms and set predicted value (MHT) as the main factor. Divided them into train and test data set (70:30).
- Implemented **classification models** such as decision tree, naive bayes, bagging, boosting and random forest. Calculated the **confusion matrix** and **AUC** followed by plotting the **ROC** curve to obtain the performance of each model and variable importance.
- Selected and omitted variables based on the variable importance and re-obtain the accuracy to evaluate the performance if any improvement is observed.
- Implemented and observe the output of two additional models, **Artificial Neural Network (ANN)** and **K-Nearest Neighbor (KNN)** to compare with previous models performance to identify the best performing model and common strong variables predictor of MHT (More Humid Tomorrow).

3. Natural Language Processing for Network Analysis

- Collected text samples of three different categories from five different sources each (Google Scholar).
- Performed **tokenization** of collected texts to remove numbers, punctuation and lower case conversion as well as filtering stop words and white space.
- Created two **document-term matrix** of pre-processed texts into csv file where one is normal and one with sparse terms (less word occurrence) removed.
- Plotted a **hierarchical clustering (dendrogram)** to identify the relations of word clusters in the collected texts (qualitative).
- Obtained the **confusion matrix** between topic labels and cluster group for accuracy (quantitative).
- Implemented **single-mode network** and bipartite **(two-mode) network** to showcase the degree, betweenness, closeness and eigenvector that identifies the relationship between each node (word or document).

SKILLS

Data Cleaning/Tidying	Descriptive Analysis	Data Analysis and Statistic
Data Visualization	Library Tool Usages for Python and R	Dashboard Automation
External Data Collection	DBMS	Data Modeling

TECHNICAL TOOLS EXPERIENCE

PROGRAMMING LANGUAGE

- Python, R, SQL, Java, Javascript, C, C++

FRAMEWORK

- Django

DATA ANALYTIC

- Microsoft: Excel, Google Sheets, Power Bi
- Python: Pandas, Numpy, Matplotlib, Plotly, NLP
- R: TidyVerse, ggplot2

AI TOOL

- Ollama, OpenAI, Gemini, Digital Ocean’s Inference, HuggingFace

---

EDUCATION & CERTIFICATIONS

Bachelor of Computer Science **July 2020 - April 2025**
Major: Advanced Computer Science
Monash University Malaysia

Monash University Foundation Year **July 2019 - June 2020**
Sunway University

GCE O-Level **January 2016 - November 2018**
Chung Ching Middle School